

Stoichiometry Road Map

Four stops on one road, and the single arrow on it that is allowed to change which substance you are talking about.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/stoichiometry-road-map/

1 Before you open anything

Answer from what you already know, before the screen tells you anything. The point is to have a number on paper to argue with later.

PREDICTION

Propane burns according to $\text{C}_3\text{H}_8 + 5 \text{O}_2 \rightarrow 3 \text{CO}_2 + 4 \text{H}_2\text{O}$. You burn **25.0 g** of propane and every gram of it reacts. Is the mass of CO_2 produced *more than, less than, or equal to* 25.0 g? Commit to an actual number of grams.

Write one or two sentences on how you got that number — including anything you assumed without checking.

2 The two arrows that stay inside one substance

Open the demonstration. The four buttons at the top of the control column choose the level; everything below them changes to match.

1. Click **1 • Molar mass**. Clear the **Chemical formula** box and type **C3H8**. Capitalisation matters — the box will tell you if it cannot read what you typed.
2. Record the **Molar mass** and **Atoms per unit** readouts, then read the *Share of mass* column of the *Atom by atom* table and record whichever element holds the largest share.
3. Repeat for the other three formulas in the table. Those three are preset buttons under the formula box; C_3H_8 is the only one you have to type.

Formula	Molar mass (g/mol)	Atoms per unit	Element with the largest share of the mass, and its %
C ₃ H ₈			
CO ₂			
Ca(OH) ₂			
Al ₂ (SO ₄) ₃			

a. Ca(OH)₂ reports 5 atoms per unit, not 3. Say exactly what the subscript 2 outside the parenthesis is doing.

b. C₃H₈ and CO₂ have molar masses that agree to within 0.1 g/mol, yet the *Atom by atom* table gives one of them 81.71% carbon by mass and the other 27.29%. Explain how both can be true.

Now the second arrow. Click **2 · Convert**.

1. Set **Substance** to **C3H8 — propane**, **You are given** to **grams**, the **Amount** box to **25.0**, and **Find** to **moles**.
2. Record the **Answer (moles)** readout.
3. Change **Find** to **particles** and record **Answer (particles)**. Count the conversion factors now on the *The work* line.

Find set to	Answer readout (with units)	How many fractions appear in <i>The work</i>
moles		
particles		

c. There is no single fraction on the screen that turns grams straight into molecules. Using the road map picture, say why moles has to sit in the middle of that route.

Under the Amount box the demonstration reports how many significant figures your entry carries and rounds the answer to match. Typing **25** instead of **25.0** changes the reported answer even though the arithmetic is identical. The atomic masses it uses are the IUPAC 2021 conventional values, and a molar mass built by addition keeps the fewest decimal places of any atomic mass in it — that is why $\text{Al}_2(\text{SO}_4)_3$ stops at two decimals while $\text{Ca}(\text{OH})_2$ keeps three.

3 The whole road, one target at a time

Click **4 • Road map**. This is the level that settles your prediction. You will change one control — **Find grams of** — and leave everything else alone.

1. Set **Reaction** to **Combustion of propane, You have grams of to C₃H₈ — propane**, and the **Grams** box to **25.0**. Leave **Show every species in the reaction** ticked.
2. Set **Find grams of** to **CO₂** and copy all four readouts into the table.
3. Repeat with **H₂O**, then with **O₂**.

Find grams of	Mole ratio, read off the boxed coefficients	Moles C ₃ H ₈	Moles of the target	Grams of the target
CO ₂				
H ₂ O				
O ₂				

a. The *Moles C₃H₈* readout never changed across the three rows, and the other two columns did. Which single quantity is responsible for each of those two changes?

b. State the rule in your own words: to get from grams of A to grams of B, what exactly do you multiply the grams of A by? Write it as one expression in terms of the two coefficients and the two molar masses.

c. Go back to your prediction in part 1. What is the actual mass of CO₂? If you were wrong, name the specific assumption that did it — not “I made a mistake”.

d. In *The whole reaction, at this scale* table, compare the **Reactants total** row with the **Products total** row. The mass cells agree and the mole cells do not. Explain why that is the expected result rather than an error.

4 Upside down, and backwards

Two things a test question will do to you: hand you the product instead of the reactant, and reward you for noticing an inverted conversion factor before the answer is wrong.

1. Click **3 • Mole ratio**. Set **Reaction** to **Haber process (ammonia)**, **You know moles of** to **H₂**, the **Moles** box to **4.00**, and **Find moles of** to **NH₃**.
2. Record the **Mole ratio** readout and the moles of NH₃.
3. Now tap the fraction in *The work – tap a fraction to flip it*. Record the number the result box now shows and what it says in place of a unit. Tap again to put it back.

Orientation of the fraction	Mole ratio readout	Result value	Units on the result
As given			
Flipped			

a. Both orientations produce a perfectly ordinary number. Describe the check that separates them, in a form you could run on paper in about five seconds.

b. Skipping the mole ratio entirely would give 4.00 mol of NH₃. What would have to be true of the balanced equation for that to be the right answer? Is it true here?

Now run the road backwards. Click **4 • Road map**.

1. Set **Reaction** to **Haber process (ammonia)**, **You have grams of** to **NH₃ – ammonia**, **Grams** to **34.0**, and **Find grams of** to **H₂**.
2. Record all four readouts.

READOUTS

Grams NH_3 _____ Moles NH_3 _____ Moles H_2 _____ Grams H_2 _____

The substance you started with is a *product* this time, so the middle arrow runs the other way. Write the mole ratio you used, with the correct quantity on top, and say which coefficient supplied each number.

Read the panel headed **What this assumes** beside the readouts. Every number this level produces assumes the reaction runs to completion at 100 percent yield, with the starting substance completely consumed. Nothing on this worksheet is a prediction of what a real flask gives you — limiting reactant and percent yield are what close that gap.

5 Solve these without the demonstration

Close the page or look away from it. Show every conversion factor as a fraction, with units. Then reopen the demonstration and check yourself — and if a number is off, find the step it went wrong at rather than just correcting the total.

1. Find the molar mass of $\text{Ca}(\text{OH})_2$, then the percentage of its mass that is oxygen. Give the molar mass to the correct number of decimal places for a sum, and the percentage to two decimal places.

2. The thermite reaction is $2 \text{Al} + \text{Fe}_2\text{O}_3 \rightarrow \text{Al}_2\text{O}_3 + 2 \text{Fe}$. What mass of iron is produced from 50.0 g of Fe_2O_3 , assuming complete reaction? Report to three significant figures.

3. How many *oxygen atoms* are in 25.0 g of $\text{Al}_2(\text{SO}_4)_3$? Note that the demonstration will report formula units, so one step of this is yours alone — state which.

4. A shortcut worth taking seriously: divide the grams of A by the molar mass of A and multiply by the molar mass of B, with nothing in between. For $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$ it gives the right mass of CO_2 every time. For $2 \text{KClO}_3 \rightarrow 2 \text{KCl} + 3 \text{O}_2$ it does not give the right mass of O_2 . Explain why it survives the first reaction, and state the exact factor by which it is wrong in the second.

6 On your own

Nothing here is graded, and none of it appeared above. The demonstration does considerably more than this worksheet uses.

- On level 1, type **CuSO4*5H2O** and then **CuSO4**. What fraction of a hydrate's mass is water, and what happens to a mass measurement if the crystals are damp?
 - Still on level 1, compare **CO** with **Co**. Decide what each one is before you look at the molar mass.
 - On level 2, cycle the Substance through **Fe**, **NaCl** and **CH4** with Find set to particles. The noun on the answer changes each time. Work out why NaCl is not counted in molecules.
 - On level 4, try **Photosynthesis (overall)** and then **Decomposition of potassium chlorate** with every species shown. Find a reaction where the total moles of product exceed the total moles of reactant, and one where they fall.
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